Lambert problem

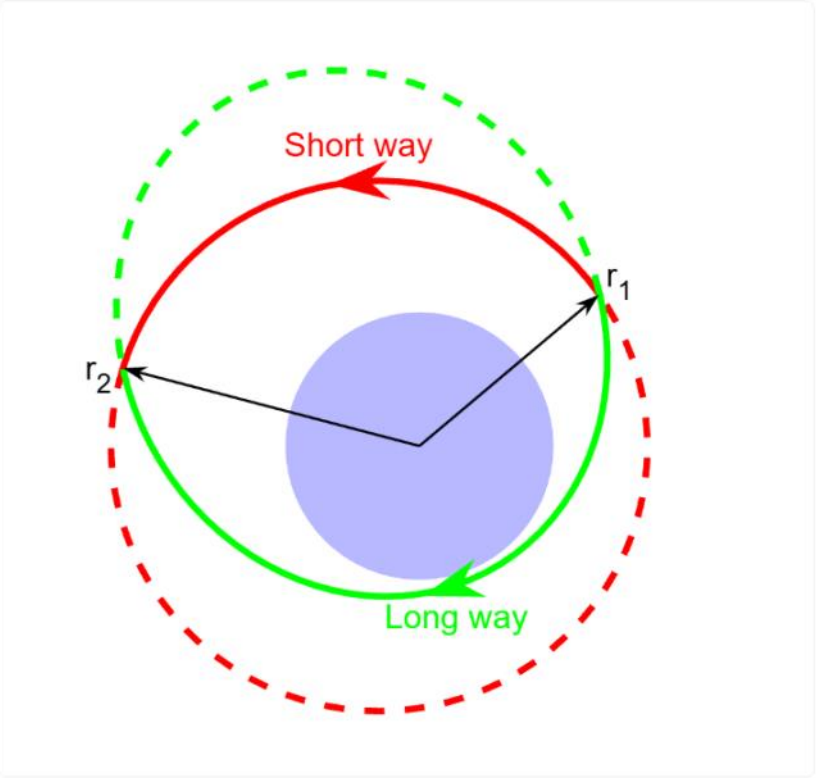
Thursday, October 8, 2026 10:24 AM

Two-point boundary value problem: given a satellite position at two different times, what is its orbit?
Most common application is for orbit determination

Curtis section 5.3, but I will use an approach discussed in the book by Prussing and Conway

Lambert set: $(\vec{r}_1, \vec{r}_2, t)$

Lambert mode or route



Suppose a satellite passes between two points

$$\boldsymbol{r}_1 = -6991\hat{\boldsymbol{I}} - 5118\hat{\boldsymbol{J}} + 7078\hat{\boldsymbol{K}}\,\text{km}$$
$$\boldsymbol{r}_2 = 11326\hat{\boldsymbol{I}} - 6317\hat{\boldsymbol{J}} - 6677\hat{\boldsymbol{K}}\,\text{km}$$

5200 seconds apart.

Short way $\Delta\theta < 180$ deg
Long way $\Delta\theta > 180$
Colinear $\Delta\theta = 180$

Solving the zero-revolution Lambert problem
Start with Kepler's equation: $M = E - e \sin E$

$$\Delta t = t_2 - t_1 = \frac{1}{n}\left[E_2 - E_1 - e(\sin E_2 - \sin E_1)\right]$$

Define $E_+ = \frac{E_2 + E_1}{2}$ $E_- = \frac{E_2 - E_1}{2}$

$$r = a(1 - e\cos E)$$

Wikipedia "product to sum and sum to product identities"

$$r_1 + r_2 = a\left[2 - e(\cos E_1 + \cos E_2)\right]$$
$$r_1 + r_2 = 2a\left[1 - e\cos E_+ \cos E_-\right]$$

$$0 \le E_1 < E_2 \le 2\pi$$

$$0 \le E_+ \le 2\pi$$
$$0 \le E_- \le \pi$$

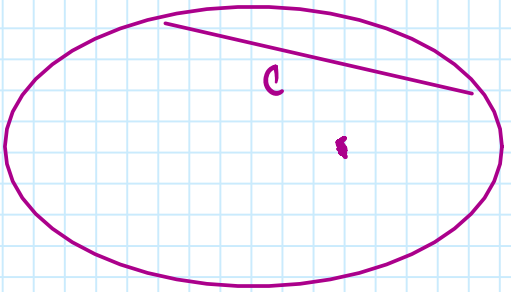
Lecture 13
10/15/2026

- HW#4 posted, due next Wednesday
- Midterm exam - see announcement
- Tellurion and file uploads - see announcement
- HW#3 - problem 4.6 answer in book is not correct

Chord

$$x = a\cos E$$

$$y = b\sin E \hookrightarrow a\sqrt{1-e^2}$$



$$c^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

$$= a^2(\cos E_2 - \cos E_1)^2 + a^2(1-e^2)(\sin E_2 - \sin E_1)^2$$

$$= 4a^2 \sin^2 E_+ \sin^2 E_- + 4a^2(1-e^2)\cos^2 E_+ \sin^2 E_-$$

$$= 4a^2 \sin^2 E_- (1 - e^2\cos^2 E_+)$$

Assume a bound orbit and introduce a new angle xi

$$e < 1 \Rightarrow |e\cos E_+| < 1$$

$$\cos\xi = e\cos E_+$$

$$0 \le \xi \le \pi$$

With xi, chord equation is

$$c = 2a \sin E_- \sin\xi$$

Then this equation

$$r_1 + r_2 = 2a\left[1 - e\cos E_+ \cos E_-\right]$$

is

$$r_1 + r_2 = 2a\left[1 - \cos E_- \cos \xi\right]$$

Define two angles as sum and difference of xi, E_

$$\alpha = \xi + E_- \qquad 0 \le \alpha \le 2\pi$$

$$\beta = \xi - E_- \qquad -\pi \le \beta \le \pi$$

$$2E_- = \underbrace{E_2 - E_1}_{\Delta E} = \alpha - \beta$$

α, β Can be expressed as the sum and differences of the radii sum and chord

$$r_1 + r_2 + c = 2a(1 - \cos E_- \cos\xi + \sin E_- \sin\xi)$$
$$= 2a(1 - \cos\alpha) = 4a\sin^2\frac{\alpha}{2}$$

$$r_1 + r_2 - c = 2a(1 - \cos E_- \cos\xi - \sin E_- \sin\xi)$$
$$= 2a(1 - \cos\beta) = 4a\sin^2\frac{\beta}{2}$$

Transfer time equation

$$e(\sin E_2 - \sin E_1) = 2e\cos E_+ \sin E_-$$
$$= 2\cos\xi \sin E_-$$

$$= 2\cos\xi \sin E_-$$
$$= \sin(\xi + E_-) - \sin(\xi - E_-)$$
$$= \sin\alpha - \sin\beta$$

$$\Delta t = \frac{1}{n}\left[\alpha - \beta - (\sin\alpha - \sin\beta)\right]$$

Lambert's theorem
The transfer time is a function of sma, sum of the two radial distances, chord length* (* = terms and conditions apply)

$$\Delta t = F(a, r_1 + r_2, c) = \frac{1}{n}\left[(\alpha - \sin\alpha) - (\beta - \sin\beta)\right]$$

$$r_1 + r_2 + c = 4a\sin^2\frac{\alpha}{2}$$

$$r_1 + r_2 - c = 4a\sin^2\frac{\beta}{2}$$

16 possibilities because of the half angles and the squares
NOOOOOO!!!!

$$0 \le \alpha < 2\pi \qquad -\pi < \beta \le 2\pi$$

Actually only 4 possibilities

$\frac{\alpha}{2}$ is in quadrants I or II

$\frac{\beta}{2}$ is in quadrants I or IV

1) Is $\cos\frac{\alpha}{2} \lessgtr 0$?

2) Is $\sin\frac{\beta}{2} \lessgtr 0$?

Long way
Short way

$\cos\frac{\alpha}{2}$

$$r_1 + r_2 + c = 4a$$

$$2a - s_1 + 2a - s_2 + c = 4a$$

$$c - s_1 - s_2 = 0$$

Excludes vacant focus:

$$\cos\frac{\alpha}{2} > 0$$

includes:

$$\cos\frac{\alpha}{2} < 0$$

Can be solved from the flight time

Given r1+r2, chord, sma - 4 possible solutions

α^*, β^* sol'ns short arc

$\alpha = 2\pi - \alpha^*$, $\beta = -\beta^*$

1) $\alpha = \alpha^*$, $\beta = \beta^*$

2) $\alpha = \alpha^*$, $\beta = -\beta^*$

3) $\alpha = 2\pi - \alpha^*$ $\beta = \beta^*$

3) $\alpha = 2\pi - \alpha^*$, $\beta = \beta^*$

4) $\alpha = 2\pi - \alpha^*$, $\beta = -\beta^*$

If

$$a < a_{min} = \frac{r_1 + r_2 + c}{4}$$

There is no possible solution, because sin^2 is not between 0 and 1

If a > amin, there are four solutions

Lecture 14
10/20/2026
Lambert solution continued

Midterm exam posted in one week, due 10/28/2026

Minimum energy or minimum semimajor solution
Occurs at $\alpha = \pi$

Substitute into Lambert function

$$\Delta t = \frac{1}{n}\left[\alpha - \beta - (\sin\alpha - \sin\beta)\right]$$

$$t_{crit} = \frac{1}{n_{crit}}\left[\pi - (\beta_{crit} - \sin\beta_{crit})\right]$$

$$\rightarrow n_c = \sqrt{\frac{\mu}{a_{min}^3}}$$

"crit" → "c"

$$\sin^2\frac{\beta_c}{2} = \frac{r_1 + r_2 - c}{r_1 + r_2 + c}$$

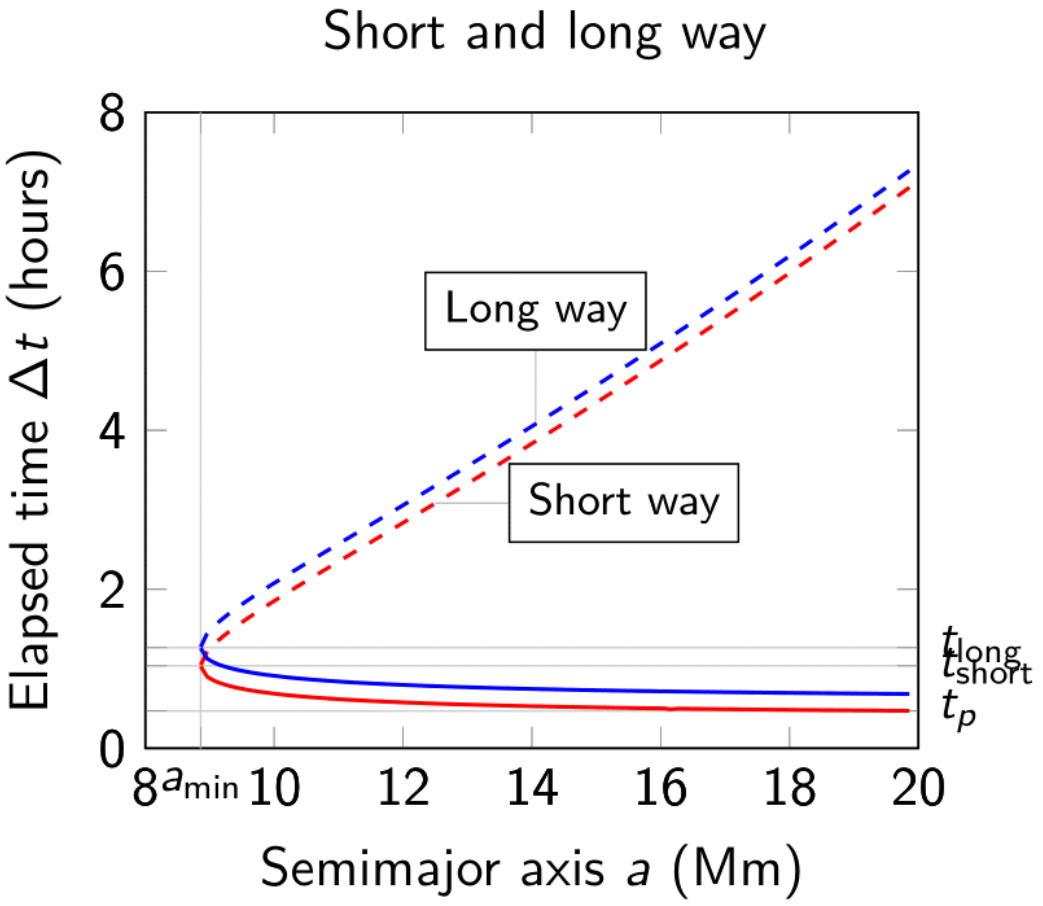
Two solutions: in quadrant I (short way), quadrant IV (long way)

$$\beta_{long} = -\beta_{short}$$

$t_c \rightarrow t_{short}$, t_{long}

Minimum time elliptic solution; Euler's equation

$$\Delta t \rightarrow t_p = \frac{1}{6\sqrt{\mu}}\left[(r_1 + r_2 + c)^{3/2} - (r_1 + r_2 - c)^{3/2}\right]$$



Lambert's theorem restated

Orbit time (time of flight) is dependent only on

- Semimajor axis a
- Sum of distances of terminal points r1 + r2
- Length of the chord joining the points, c
- Direction: short way or long way

$$\Delta t = F\left(a, r_1 + r_2, c, \begin{Bmatrix}\text{short}\\ \text{long}\end{Bmatrix}\right)$$

$|\vec{r}_1| + |\vec{r}_2|$ $\qquad$ $|\vec{r}_1 - \vec{r}_2|$

Lambert problem: Given that a satellite went from r1 to r2 over a time delta t, find the semimajor axis of the orbit and change the change in eccentric anomaly on that orbit between the points. (*Also, whether it went short or long way)

Iterative solution

1. Choose a value of the semimajor axis and compute α^*, β^*
2. Compute beta from beta*

$$\Delta\theta < \pi \quad \beta = \beta^*$$

$$\Delta\theta > \pi \quad \beta = -\beta^*$$

3. Compute tc

$$\Delta t < t_c \quad \alpha = \alpha^*$$

$$\Delta t > t_c \quad \alpha = 2\pi - \alpha^*$$

$$\Delta t = t_c \quad \alpha = \pi$$

4. Solve for delta t

$$\Delta t = \frac{1}{n}\left[\alpha - \beta - (\sin\alpha - \sin\beta)\right]$$

5. Iterate with new value of a until solved delta t matches actual
6. Output: semimajor axis and change in eccentric anomaly

$$a, \quad \Delta E = \alpha - \beta$$

7. Find the velocity with the Lagrange coefficients expressed in eccentric anomaly

$$f(a, r_1, \Delta E)$$

$$g(a, \Delta t, \Delta E)$$

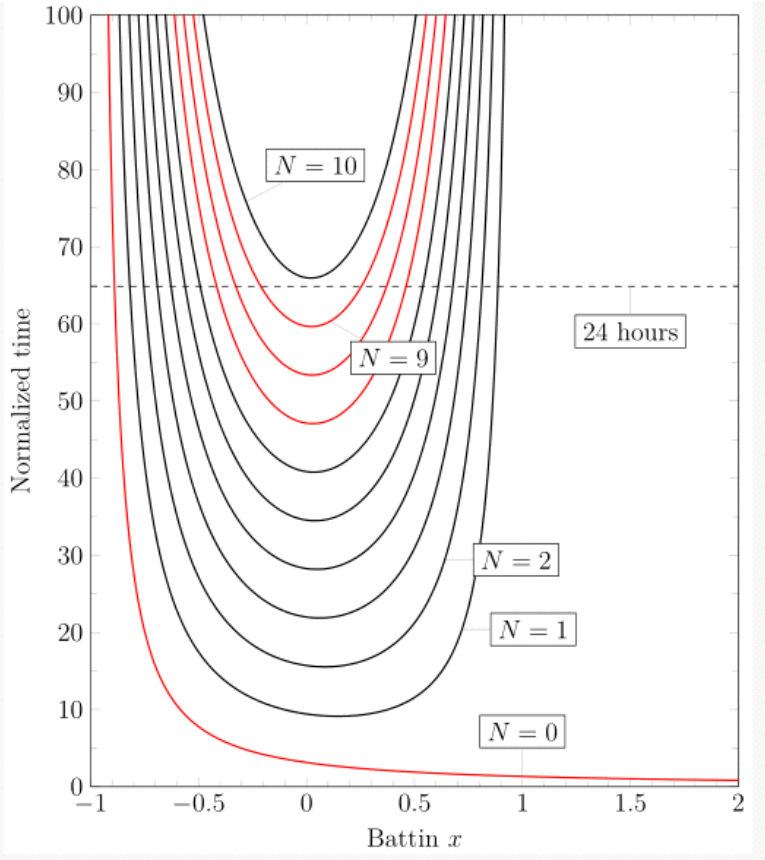
$$\vec{v}_1 = \frac{\vec{r}_2 - f\,\vec{r}_1}{g}$$

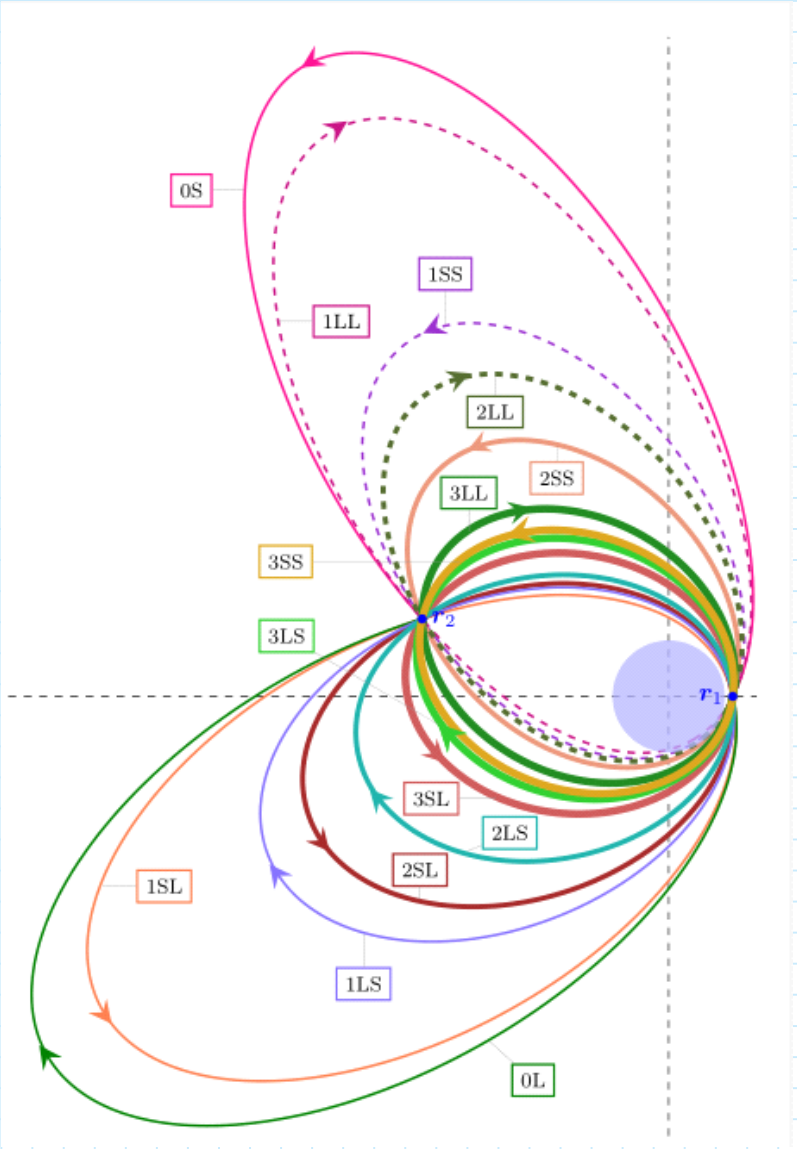
Initial velocity

Multi-revolution Lambert problem
Complete Lambert problem

- Number of whole orbits that completed before reaching r2, N
- Short/way long way (direction)
- Two solutions for each direction when N>0

There are 2 + 4Nmax solutions possible


100
90
80
70
60
50
40
30
20
10
0
Normalized time
−1
−0.5
0
0.5
1
1.5
2
Battin x
$N = 10$
$N = 9$
$N = 2$
$N = 1$
$N = 0$
24 hours



Lambert algorithms
Survey paper 2015 Sangra & Fantino
https://issfd.org/2015/files/downloads/papers/028_Sangra.pdf
Categorize by iteration variable

- Universal variables x, z
  - Lancaster & Blanchard (1969), Bate (1971), Gooding (1990), Izzo (2015)
  - Elliptical, parabolic, hyperbolic
- Semimajor a
  - Lagrange, Thorne, Prussing
  - Classic approach
  - Two arms - decide which
  - Iteration solver may have trouble solving near amin.
- Semilatus rectum (p)

    - Herrick-Liu, Boltz
    - "p-iteration"
- Kustaanheimo-Stiefel (K-S) regularized coordinates
    - Simo, Jezewski

Newer algorithms include Russell's iterated vercosine, which is a complete multi-rev solver

1. Russell, Ryan P. “On the Solution to Every Lambert Problem.” Celestial Mechanics and Dynamical Astronomy 131, no. 11 (2019): 50. doi:/10.1007/s10569-019-9927-z.
2. Russell, Ryan P. “Complete Lambert Solver Including Second-Order Sensitivities.” Journal of Guidance, Control, and Dynamics 45, no. 2 (2022): 196–212. doi:/10.2514/1.G006089.

Russell iterates on

$$k^2 \equiv \operatorname{vercos}\left(\frac{x}{\sqrt{a}}\right) \equiv 1 + \cos\left(\frac{x}{\sqrt{a}}\right)$$

Where x is a universal variable from the Sundmann transformation (form of regularization)

Precompute and store tables of coefficients, Russell supplies the code (Fortran and Matlab) and the tables.

Feature	Russell 2019 (ivlam)	Russell 2022 (`ivlam2`)
Interpolation	Biquintic splines on a 2D rectangular grid.	Low-order multivariate polynomials on KD trees.
Memory	~3 to 65 MB per revolution.	Single 1 MB coefficient file for any N.
Parameter Space	Solved separately for each revolution.	Uses transformations to make surfaces nearly invariant to N.
Sensitivities	Not explicitly included in the base solver.	Exact 1st and 2nd-order partial derivatives.